

TECHNICAL RESEARCH REPORT

Multilingual NLP *Architectures* & Their Linguistic Foundations

*A comparative analysis of mBERT, XLM-R, ByT5, and GPT,
and the question of whether statistical language models can be
said to understand.*

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Tokenization & the Multilingual Transformer

Most modern NLP architectures share a common move: they break language into statistical fragments and learn from how those fragments co-occur. The differences between them lie almost entirely in how they fragment, and those differences turn out to be deeply linguistic.

mBERT WORDPIECE

mBERT is trained on Wikipedia articles across 104 languages, and crucially it treats all of them as a **single language**. It achieves this through a shared WordPiece vocabulary: a dictionary common to every language in the corpus.

WordPiece performs **subword tokenization**: it splits sentences and words into smaller pieces and ranks those pieces by how frequently they appear adjacent to one another in the corpus. From a linguistic standpoint, it segments words into morpheme-like units **without distinguishing bound from free morphemes**, and uses adjacency frequency to decide which units belong together. This lets the model represent rare or unseen words as a combination of known subwords.

Consider the Turkish word `gözlükçü` ("optician"). It decomposes into `göz` (eye), `-lük`, and `-çü`. WordPiece observes each piece in isolation and in combination, recognizes that `göz` stands alone as a noun, notices that `-lük` and `-çü` recur alongside nouns like it, and adds these patterns to its vocabulary with the relevant parameters.

The algorithm works hierarchically, preferring the longest available subwords and falling back toward shorter ones. The result is effectively a **morphology-driven tokenizer**, and that morphological grounding arguably places mBERT ahead of many alternatives for structurally rich languages.

THE LIMITATION

This kind of algorithm is sufficient for text generation and similar tasks. But the moment it must hold a dialogue or accomplish something through genuine back-and-forth conversation, it falters: because it encodes no information about **context or meaning**, it cannot reliably know how to match expressions whose sense isn't recoverable from their morphemes alone.

PRE-TRAINING PROCESS

The model is exposed to text in multiple languages and encodes semantically similar words as nearby vectors. It then continuously updates and extends the WordPiece set through subword tokenization. Vectorization can use parallel texts (labeled data) that are translations of one another, but via VecMap it can also be trained on unlabeled data through matrix operations.

FINE-TUNING PROCESS

The pre-trained model is then optimized for specific tasks (sentiment analysis, chat, etc.) using labeled data from a high-resource language such as English.

TRANSFER PROCESS

Because pre-training encodes all languages into shared semantic vectors, mBERT can perform in a language it was never explicitly fine-tuned on, at a level approaching its training language. These direct transfers are called *zero-shot transfers*. In *few-shot transfer*, a small amount of labeled data in the target language helps the model grasp that language's nuances more precisely.

POS TAGS & DEPENDENCY STRUCTURES

Universal linguistic properties are encoded so that models learn to recognize them during pre-training. Take the Turkish sentence "Ali ile dün sinemaya gittik" ("We went to the cinema with Ali yesterday"):

Ali → **Subject** ile → **Conjunction** dün → **Adverb**
sinemaya → **Object** gitti(k) → **Verb** (-k) → **Embedded subject**

By encoding structure this way, the model can match words with similar encodings across languages. **Dependency structures** push this further: by examining a sentence's logical relation graph, structures with similar logical relations are encoded close together in vector space.

EN He (X) killed (Y) the guy (Z)
TR O (X) adamı (Z) öldürdü (Y)

The agent (X), the action (Y), and the affected entity (Z) map onto one another across languages despite differing word order.

ADVERSARIAL LEARNING

The model is split into a **Generator** and a **Discriminator**. After training, the system is deliberately challenged: the Generator produces data the model struggles to classify, while the Discriminator tries to classify it correctly. The goal is to minimize the resulting objective:

$$\min_{\theta} \max_{\delta \in S} L(f(x + \delta ; \theta), y)$$

PSEUDO-LABELING

Predicted labels are assigned to the target language's unlabeled data, allowing the model to be fine-tuned on them.

mBERT is publicly available and runs on any local or cloud system (Google Colab, AWS) with the relevant Python libraries installed. The HuggingFace Transformers library is the most common entry point. It has no speech-to-text capability. The sections below show three stages in code: loading the model and generating embeddings, tokenizing across multiple languages, and fine-tuning for a specific task.

LOADING THE MODEL & GENERATING EMBEDDINGS

The model and tokenizer load directly from `bert-base-multilingual-cased`. A sentence is tokenized, converted to a tensor, and passed through the model to obtain its last hidden states.

```
# Load the mBERT model and tokenizer
import torch
from transformers import BertTokenizer, BertModel

model = BertModel.from_pretrained('bert-base-multilingual-cased')
tokenizer = BertTokenizer.from_pretrained('bert-base-multilingual-cased')

# Tokenize, encode, and generate embeddings
input_ids = torch.tensor([tokenizer.encode("This is an example sentence.",
add_special_tokens=True)])
with torch.no_grad():
    outputs = model(input_ids)
last_hidden_states = outputs.last_hidden_state
print("Last Hidden States Shape:", last_hidden_states.shape)
```

TOKENIZING & EMBEDDING ACROSS LANGUAGES

Because mBERT treats all languages through one shared vocabulary, the same loop produces embeddings for English, Chinese, French, and Spanish alike.

```
sentences = [
    "This is an example sentence.",      # English
    "这是一个例句。",                    # Chinese
    "Ceci est une phrase exemple.",      # French
    "Esto es una oración de ejemplo."    # Spanish
]

for sentence in sentences:
    input_ids = torch.tensor([tokenizer.encode(sentence, add_special_tokens=True)])
    with torch.no_grad():
        outputs = model(input_ids)
    print(f"Sentence: {sentence}", outputs.last_hidden_state.shape)
```

FINE-TUNING FOR A SPECIFIC TASK

For classification, the model is loaded with a task-specific head (`BertForSequenceClassification`); a forward pass with labels returns both the loss and the logits.

```

from transformers import BertForSequenceClassification
model = BertForSequenceClassification.from_pretrained(
    'bert-base-multilingual-cased', num_labels=2)

input_ids = torch.tensor([tokenizer.encode("This is an example.",
add_special_tokens=True)])
labels = torch.tensor([1]) # dummy binary label
outputs = model(input_ids, labels=labels)
loss, logits = outputs.loss, outputs.logits

```

XLM-R Large BPE

Like mBERT, XLM-R operates through a single shared transformer architecture, enabling knowledge learned in one language to transfer to another. But unlike mBERT, it uses **Byte-Pair Encoding (BPE)** rather than WordPiece for tokenization.

Both methods fundamentally split words into morpheme-like pieces and rely on a mathematical convergence function. The difference is subtle but important: WordPiece decides which pairs to merge using a *similarity* heuristic tied to its training data, whereas **BPE works purely quantitatively**, directly counting how often morphemes occur adjacent to one another. The vocabulary is updated by greedily merging the pair that maximizes the objective over the candidate set s :

$$\text{Vocabulary Update: } V_{\text{new}} = \text{Merge}(\arg \max_s L(s))$$

XLM-R is trained on far more data than mBERT and runs with roughly **twice the parameters** (24 layers, 1024 hidden units). This makes it faster, more accurate on low-resource languages, and generally higher-performing. Like mBERT, it uses Masked Language Modeling during pre-training, masking around 15% of input tokens and predicting them from the surrounding context, which strengthens its contextual representation and improves performance on context-dependent communication. It too lacks speech-to-text.

```

from transformers import XLMRobertaTokenizer, XLMRobertaForSequenceClassification
import torch

tokenizer = XLMRobertaTokenizer.from_pretrained('xlm-roberta-base')
model = XLMRobertaForSequenceClassification.from_pretrained('xlm-roberta-base')

inputs = tokenizer("This is a test sentence.", return_tensors="pt")
outputs = model(**inputs)
logits = outputs.logits

```

ByT5 BYTE-LEVEL

Unlike the others, ByT5 does not tokenize text into morphemes at all. Instead of splitting a structure into pieces to represent its morphemes, it works **directly at the byte level**, using the byte sequence that represents the structure.

This sidesteps the faulty-tokenization problem that arises in morphologically complex languages. Its byte-level processing power gives it a second advantage: strong performance across many languages at once. Because its core algorithm isn't built on any particular language, ByT5 occupies the position of a genuinely **universal NLP model**. Like its predecessor T5, it runs on an encoder-decoder architecture.

It handles translation, classification, question-answering, and summarization for any language without language-specific pre-processing. However, it has no speech-to-text, and byte-level processing is more expensive than tokenization, so despite its advantages, it can become disadvantageous depending on a project's structure, cost, and expected return.

```
from transformers import ByT5Tokenizer, ByT5ForConditionalGeneration

tokenizer = ByT5Tokenizer.from_pretrained("google/byt5-small")
model = ByT5ForConditionalGeneration.from_pretrained("google/byt5-small")

inputs = tokenizer("Translate English to French: How are you?",
return_tensors="pt")
outputs = model.generate(**inputs)
decoded = tokenizer.decode(outputs[0], skip_special_tokens=True)
```

GPT BPE VARIANT • ATTENTION

GPT uses a variant of BPE for tokenization, matching the most frequently adjacent morphemes by raw frequency. What sets it apart from other models using the same tokenization method is its **architecture**.

During processing, GPT computes multiple weights, splitting each word into three vectors: **Query (Q), Key (K), and Value (V)**. For every pair W_x and W_y , it computes an attention score:

$$\text{Attention Score}_{ij} = Q_i \cdot K_j^T / \sqrt{d_k}$$

The resulting score is squeezed between 0 and 1 via a softmax probability calculation, and this process is applied across the whole sequence $W \cdot W_{(n1)} \cdot W_{(n2)} \dots W_{(nn)}$.

After computing the proximity relations between ordered morphemes far more efficiently, there's a catch: because attention scores are computed in a distributed way, the self-attention mechanism **cannot directly access positional information**. To solve this, **positional encoding** distributes position information back to the relevant W 's.

In computing self-attention scores, GPT does **not** proceed sequentially like RNN architectures: it computes simultaneously and in parallel. This buys speed and optimization, but at a cost: some parametric information cannot be computed alongside the rest. These packaged representations are then passed to the relevant neural networks, and weights and biases are distributed.

```
import openai
openai.api_key = 'your-api-key-here'

response = openai.ChatCompletion.create(
    model="gpt-4",
    messages=[
        {"role": "system", "content": "You are a helpful assistant."},
        {"role": "user", "content": "How can I use the ChatGPT API?"},
    ]
)
print(response['choices'][0]['message']['content'])
```

COMPARATIVE SUMMARY

mBERT: morphology-driven (WordPiece); strong structure, weak context. **XLM-R**: quantitative BPE, $2\times$ parameters; faster and stronger on low-resource languages. **ByT5**: byte-level, language-agnostic, universal but costly. **GPT**: BPE variant distinguished by parallel self-attention with positional encoding.

On the Architecture of Current NLP Models

Almost every model tokenizes language through a morphological lens. A language's formal structure can encode a great deal of structural and semantic information, but it cannot, on its own, encode meaning and context. The reason is foundational: language is arbitrary.

There is no discoverable reason why the word `masa` ("table") represents an actual table in the world; none exists, and none can be found. The mind encodes the conceptual content of the world's projections first, and then maps the sounds that represent them. But this production-and-matching process is **entirely arbitrary and non-correlative**; it carries no meaningful correlation.

This is precisely why correlation-driven morpheme probes like BPE can work so efficiently: the correlation between words and their meanings is not *semantic* but **usage-based**. The models latch onto patterns of use, not patterns of meaning.

THE CORE CLAIM

Current models are powerful at detecting word-level relations. But the underlying operation is a **statistical analysis of data**. With the present epistemological method (optimizing statistical functions, summing many linear equations into neural networks), it does not appear possible to obtain an artificial "mind" that genuinely reasons, that answers *because it understands*.

What we are likely to reach instead is a system that *speaks like* a human mind without ever moving beyond that imitation. (cf. Searle's Chinese Room.)

Human & Machine: Acquisition vs. Learning

A newborn acquires its first language solely by being exposed to the linguistic input around it. In the absence of such environmental input, a person ends up with profoundly impaired cognitive function. The newborn acquires its first language **directly**: building its parameters straight from the language, thinking in it, and gaining productive fluency remarkably fast. But once the first language is acquired and the child passes one or two years of age, neuroplasticity drops sharply: from that point on, languages are no longer *acquired*; they are *learned*.

The fundamental difference between learning and acquisition is one of **principles and parameters**. Every human is born with the innate capacity to acquire human language (any language) with extraordinary speed and efficiency, using an innate *Language Acquisition Device*. This device contains structures called **principles**: the ability to form logical relations, to distinguish subject / object / predicate, to produce within a universal linguistic structure.

In this sense, one could say the brain does not tokenize languages: it processes their **raw representations directly**, much closer in spirit to ByT5 than to a morpheme tokenizer.

A UNIVERSAL CONSTRAINT ON INFORMATION

The universal structure of language holds that no sentence can carry more than **three core members**. To this day, no human language has ever been found with a sentence structure capable of carrying more than three units of information.

```
1 member  "Aslı yaşlandı." → X: Aslı = the one aging
2 members "Aslı köpeği seviyor." → X: Aslı = lover, Y: köpek = loved
3 members "Aslı kitabı Ali'ye verdi." → X: Aslı = giver, Y: kitap = given, Z: Ali = receiver
```

All linguistic information reduces to at most three, at least one, core informational members. The human mind has encoded its languages this way because of the universe's fundamental scheme of information transfer. Machines, lacking any such informatic perspective, have not had this built into their architectures.

When the human mind begins learning a **second** language, it starts from the parameters already acquired from the native tongue. A Turkish speaker learning French, for instance, struggles with articles: Turkish has no gendered articles and only a single person pronoun where French splits across forms (*he / she / it* → *o*; *le / la / les*). Likewise, because Turkish places adjectives before nouns, adapting to French noun+adjective order takes time, and the speaker performs a syntactic **parameter transformation** from Turkish while speaking:

```
✓ Elle lit un livre rouge (correct transformation)
× "Elle rouge un livre lit." (faulty parameter transformation)
```

Here the human mind resembles the **few-shot learning** method in NLP. When learning a new language, it fuses the information carried over from the existing language with the minimal information it has from the target language, and produces output from that combination.

While neuroplasticity is high, the mind trains itself through **unsupervised learning**; as plasticity declines (and for motivational and social reasons such as compulsory foreign-language education at school or lessons taken for personal goals), it can also acquire a language through **supervised learning**. But at that stage this is no longer a *training* process so much as a **fine-tuning** one, since the necessary parameters and weight/bias values have already been formed beforehand.

THE ARCHITECTURAL PARALLEL

In this respect, multilingual NLP models that learn a language through **supervised fine-tuning** can show a strong architectural resemblance to the human mind: in both cases a foundation built once is later adapted, not rebuilt, to accommodate a new language.

Conclusion

The human mind uses input frequency as a criterion during acquisition, just as NLP models do, but over time it shifts its learning algorithm away from frequency and toward parameters that are uniquely its own.

Those parameters are formed by social and individual projections: meaning, culture, context. And so, unlike NLP models, the mind builds far more complex and chaotic computations on top of purely statistical equations, producing a generative system that stays perpetually current.

THE TAKEAWAY

If we want a model that genuinely surpasses today's, simply making existing architectures **more complex will not be the ultimate solution**. That path raises cost while only making the imitation "better." The bottleneck is not scale: it is the absence of an informatic perspective on meaning that the human mind possesses by design.

GLOSSARY

Morphology: The study of the structure of a language or system; one of the six core fields of linguistics.

Morpheme: The smallest meaningful unit. E.g. *Sular kesildi* = su (noun) · -lar (plural) · kes (verb) · -il (passive) · -di (past tense).

Probe: A linguistic unit programmed to match a specific syntactic structure in the mind.

Language Acquisition Device: The innate language-acquisition algorithm the human mind is born with.

Written in August 2024 during a research internship. The model landscape reflects that time; the linguistic and philosophical arguments in Part II are not time-bound. Technical terminology is retained in English throughout, as in the original.